

MSPs, Networks, Lock-In, Monopolies & Adoption: Recap + Cases

What we covered

- **1. Fundamentals of Digital Economy**
- Digitization → Digitalization → Digital Transformation
- Zero marginal cost
- API economy
- Cloud, data, algorithms
- Economics of information goods
- **2. Multi-Sided Platforms (MSP)**
- Many users on two+ sides
- Cross-side interactions
- Chicken-and-egg problem
- Platform pricing & governance

What we covered contd.

- **3. Network Effects & Positive Feedback**
- Direct & indirect effects
- Data network effects
- Local vs global effects
- Negative network effects (congestion)
- **4. Lock-In & Switching Costs**
- Habit & learning curve
- Data portability issues
- Walled gardens
- Multi-homing vs single-homing

What we covered contd.

- **5. Formation of Digital Monopolies**
 - Economies of scale
 - Economies of scope
 - Zero marginal cost
 - Entry barriers
 - Winner-take-most markets
- **6. Measuring Digital Adoption**
 - World Bank's Digital Adoption Index (DAI)
 - OECD Digital Government Index (DGI)
 - Nepal, India, Sri Lanka context

Fundamentals

- **Digital economy = information + connectivity + computation**
- Value comes from data, users, algorithms
- Products = infinitely replicable
- Cost of serving next user = near zero
- **Transaction cost reduction**
- Search cost ↓ (Google)
- Coordination cost ↓ (Uber, Pathao)
- Payment cost ↓ (UPI, eSewa)
- **Example**
- → You don't search for a taxi physically → transaction cost reduced by platform.

Multi-Sided Platforms (MSP)

- **Characteristics**

- Two or more user groups
- Platform provides tools to interact
- Network effects drive value

- **Examples**

- Ride-hailing: Drivers ↔ Riders
- Food delivery: Restaurants ↔ Couriers ↔ Customers
- Online markets: Sellers ↔ Buyers
- Social media: Creators ↔ Viewers ↔ Advertisers

MSP Challenges

- **Chicken-and-egg problem**
- How to attract both sides at launch?
- Solutions: subsidies, guarantees, influencers
- **2. Disintermediation**
- User tries to bypass the platform
- Example: Customer contacts Pathao driver directly
- **3. Platform envelopment**
- Big platforms invade smaller ones
- Meta copying TikTok → Reels
- Google integrating shopping in Search

Network Effects

- More users = more value
- Direct vs indirect effects
- Examples:
- Pathao (riders ↔ drivers)
- Daraz (buyers ↔ sellers)
- Leads to **positive feedback loops** → platform growth

Flywheels & Growth Loops

- Acquisition → Activation → Engagement → Retention
- Users generate more value → attracts more users
- Example: TikTok algorithm + content creation loop
- Key point: Designed, not accidental

Lock-In & Switching Costs

- Users face high cost to leave
- Types: data, habit, ecosystem, psychological
- Examples:
 - Apple ecosystem (iPhone + iCloud + Apple Music)
 - Spotify playlists → user stickiness
 - Gmail & Google Drive ecosystem
 - WhatsApp groups & chats

Digital Monopolies

- Winner-take-most, not always 100% monopoly
- Dominance factors:
 - Network effects
 - Data advantage
 - Zero marginal cost
 - Ecosystem bundling
- Examples: WhatsApp in India/Nepal, Google Search

Winner-Take-Most Markets

- **Why digital markets create giants**
- Scale advantage
- Data advantage
- Ecosystem advantage
- Zero marginal cost
- **Examples:**
- Google Search → 90% market share
- Facebook & Instagram → dominate attention
- Amazon → dominates online retail

Market Tipping

- Small platforms rarely survive once big player dominates
- Analogy: sports tournament → only 1–2 top teams reach finals
- Example: MySpace → lost to Facebook, TikTok → Reels competition

Regulation & Policy

- Why governments care: reduce monopoly risk, protect privacy, maintain competition
- **Tools:**
- Data portability
- Interoperability (able to work with each other)
- Antitrust cases
- Examples: EU vs Google, Apple; India vs TikTok ban

Digital Adoption Matters

- Adoption levels affect platform success
- DAI pillars: People, Business, Government
- DGI dimensions: User-centricity, Digital by Design, Data-driven, Proactive, Open by Default
- Example: Nepal (people strong, business moderate, government early-stage)

Case Study: Nepal's Digital Journey

- **Strengths**
- Mobile adoption high
- eSewa, Khalti widespread
- Pathao, Foodmandu, Daraz introduced platform thinking
- **Weaknesses**
- Government digital services weak
- Low trust in official platforms
- Slow interoperability
- Legacy systems & paperwork
- **Opportunities**
- National digital ID
- Public digital platforms (like India's DPI)

Case Study: India's Digital Public Infrastructure

- **UPI**
 - Eliminated transaction barriers
 - Global benchmark for digital payments
- **Aadhaar**
 - Digital identity backbone
- **ONDC (Open network for digital commerce)**
 - Marketplace for all sellers
 - Aims to reduce Amazon/Flipkart dominance

Case Study: Sri Lanka (PickMe)

- **PickMe success**
- **1. Local Adaptation**
- PickMe understood Sri Lankan realities better than any global company:
- Local traffic patterns
- Popular routes
- Payment habits (cash-first culture)
- Local language & customer support
 - It adapted its app and services to match how Sri Lankans actually travel.

Case Study: Sri Lanka (PickMe)

- **2. Cheaper & More Reliable**

- PickMe offered:
 - Lower prices for daily trips
 - More predictable fares
 - A large local driver network (so rides were easier to find)
 - Better handling of short-distance trips that global apps often ignore
- This made the platform feel more “for the people.”

Case Study: Sri Lanka (PickMe)

- **3. Multi-Service Ecosystem**

- PickMe didn't stay limited to taxis. It added:

- **Food delivery**

- **Grocery delivery**

- **Parcel delivery**

- **Logistics services**

This increased convenience and kept users inside their ecosystem, creating network effects and stronger lock-in.

Case Study: Sri Lanka (PickMe)

- **The Big Lesson**
- **Local platforms *can* beat global companies**
→ if they understand local needs better and solve real, everyday problems.
- Uber had global tech and big money.
But PickMe had:
- Local knowledge
- Local trust
- Local customization
- Local partnerships
- **In digital markets, winning is not only about size — it's about relevance.**

Connecting the Dots

- MSPs start → grow → lock users → dominate → risk regulation
- Network effects + lock-in → monopolistic tendencies
- Digital adoption determines speed & scale of dominance
- Policy ensures fair competition & citizen protection

Student Activity Instructions

Choose one platform:

- Pathao
- TikTok
- Daraz
- eSewa
- Foodmandu
- WhatsApp

Analyze using 4 concepts:

- Network effect
- Lock-in
- Switching cost
- Monopoly risk

PATHAO — (Ride-hailing Platform)

Network Effects

- More riders → more income → more drivers join → faster rides → more riders
- Peak hours = stronger network effects
- Localized network effect (works best in Kathmandu/Pokhara)

Lock-In

- Saved locations, ride history
- Familiar drivers & safety trust
- Habit of using Pathao instead of calling taxis

Switching Costs

- Low financial cost, but high convenience cost
- Learning new app (InDrive, Tootle)
- Payment integration (eSewa/Khalti in Pathao)

Monopoly Risk

- Medium
 - Competitors exist (InDrive, Tootle)
 - Pathao strongest brand but not impossible to replace
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TIKTOK — (Short-video Platform)

Network Effects

- More creators → more content → more viewers → more creators
- Algorithm improves with more users
- Global network effect, not just local

Lock-In

- Personalized feed (“For You Page”)
- Users invested in followers, drafts, content
- Creators rely on TikTok for income & reach

Switching Costs

- High for creators (audience loss)
- Medium for viewers (losing personalized recommendations)
- Time invested in curation

Monopoly Risk

- High
 - Strongest recommendation algorithm in the world
 - Instagram Reels, YouTube Shorts exist, but TikTok still dominant
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DARAZ — (E-commerce Marketplace)

Network Effects

- More sellers → more product variety → more buyers → more sellers
- Better logistics → more trust → more buyers

Lock-In

- Daraz wallet, vouchers, reward points
- Delivery tracking + customer service
- Seller ratings & reviews

Switching Costs

- Sellers depend on Daraz traffic
- Buyers lose wishlist, coupons, history
- Learning curve for new platforms

Monopoly Risk

- Medium–High
- Weak competitors (SastoDeal, HamroBazaar not as structured)
- Logistics advantage gives Daraz edge

eSewa — (Digital Wallet)

Network Effects

- More merchants accept eSewa → more users adopt → motivates more merchants
- P2P transfers (send money to friends) strengthens the network

Lock-In

- Linked bank accounts
- Recurring payments (internet, electricity, mobile top-up)
- QR payment history & saved templates

Switching Costs

- Users must re-add bank accounts in a new wallet
- Hard to shift because friends/businesses also use eSewa
- Habit + trust in one wallet

Monopoly Risk

- High
 - Dominant market share in Nepal
 - Khalti is strong competitor but far smaller
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FOODMANDU — (Food Delivery Platform)

Network Effects

- More restaurants → more customers → more orders → more restaurants
- Delivery coverage increases demand

Lock-In

- Saved addresses & order history
- Exclusive restaurant partnerships
- Personalized recommendations

Switching Costs

- Low, because users can use Pathao Food or Bhojdeals
- Restaurants and delivery prices may vary

Monopoly Risk

- Low–Medium
 - Strong competition from Pathao Food
 - Delivery fee differences influence switching
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WHATSAPP — (Messaging Platform)

Network Effects

- Classic example of strong direct network effect
- Everyone uses it → you must use it
- Groups, communities, broadcast lists

Lock-In

- All chats, groups, media stored inside WhatsApp
- Family & work groups create emotional + social lock-in
- Calls + payments (in some countries)

Switching Costs

- Extremely high
- Users would lose:
 - Groups
 - Chat history
 - Contacts
- Hard to convince all friends/family to shift

Monopoly Risk

- Very high
 - WhatsApp is dominant in South Asia, Latin America, Africa
 - Telegram/Signal exist, but far smaller user base
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Summary Table

Platform	Network Effect	Lock-In	Switching Cost	Monopoly Risk
Pathao	Medium–Strong (local)	Medium	Medium	Medium
TikTok	Very strong (global)	High	High	High
Daraz	Strong	Medium	Medium	Medium–High
eSewa	Strong	High	High	High
Foodmandu	Medium	Medium	Low–Medium	Low–Medium
WhatsApp	Very strong	Very high	Very high	Very high

Summary & Takeaways

- Digital platforms dominate via network effects, data, and scale
- Lock-in ensures stickiness & retention
- Monopolistic tendencies are natural in MSPs
- Adoption levels and government policy shape long-term outcomes
- Case studies in South Asia illustrate real-world application

Closing & Reflection

- “Every app you use is part of a complex ecosystem”
- “Understanding these dynamics prepares you for strategic thinking, innovation, and policymaking”